Nikhil Ganpat Navghade

SYSTEM APPLICATIONS ENGINEER | RF & SEMICONDUCTOR SYSTEMS | CUSTOMER ENGINEERING

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PROFESSIONAL SUMMARY

System Applications / RF Engineer with 12+ years of experience across semiconductor products, RF systems, radar SoCs, embedded systems, DSP, system integration, validation, and customer-facing engineering.

Experienced in translating customer and system requirements into technical solutions, supporting semiconductor product evaluation and integration, performing system bring-up and laboratory validation, and troubleshooting complex interactions across RF, mixed-signal, hardware, software, and DSP domains.

Strong background in RF signal chains, radar SoCs, ADC/DAC architectures, MATLAB/Simulink, Python, Embedded C, automated validation, RF characterization, and laboratory debugging. Proven experience collaborating with customers, applications, R&D;, product, and marketing teams.

CORE COMPETENCIES

- System Applications & Customer Engineering: System architecture | Customer requirements | Product evaluation | Integration & bring-up | Validation | Technical support | Root-cause analysis
- RF & Semiconductor Systems: Radar SoCs | RF TX/RX chains | ADC/DAC | RF characterization | Gain/noise/dynamic range | Linearity & spurious analysis
- Software & Tools: Python | MATLAB/Simulink | Embedded C | Java | C++ (basic) | Git | Automated validation | Test frameworks
- Signal Processing: FMCW & pulsed radar | FFT | Digital filtering | Doppler | MIMO | Beamforming | CFAR | Detection | Angle estimation
- Laboratory: Oscilloscopes | Signal generators | Spectrum analyzers | Power meters | Attenuators | Frequency multipliers

PROFESSIONAL EXPERIENCE

NXP Semiconductors — System Integration & Validation Engineer

via ACONEXT / Hays | Munich, Germany | 2026 – Present

- Perform system- and board-level validation of semiconductor radar products, from laboratory characterization through system qualification.
- Translate product requirements into validation strategies, automated test cases, measurement procedures, and performance metrics.
- Develop Python and Java automation frameworks covering 100+ validation scenarios, improving repeatability and efficiency.
- Integrate signal generators, attenuators, frequency multipliers, power meters, and other laboratory equipment into automated validation environments.
- Perform RF and system characterization using laboratory measurements and MATLAB-based analysis.
- Diagnose RF and system-level anomalies by correlating device behavior, requirements, measurements, and software analysis.
- Perform hardware/software integration and system debugging and communicate findings to cross-functional engineering teams.

Calterah GmbH — FAE & Technical Sales

Munich, Germany | Sept 2024 – Feb 2026

- Acted as a technical advisor to customers throughout the radar SoC lifecycle, from architecture and product selection through integration, validation, troubleshooting, and deployment.
- Supported customer architecture decisions involving RF front-end behavior, ADC sampling, DSP processing, interfaces, and end-to-end system performance.
- Gathered customer requirements, evaluated technical solutions, and guided successful semiconductor product integration.
- Diagnosed complex RF, mixed-signal, hardware/software, and DSP interaction issues, coordinating investigations between customers and internal R&D.;
- Supported customer system bring-up, integration, validation, and performance optimization.
- Led root-cause analysis of issues involving secure boot, OTP programming, functional-safety configuration, and Ethernet interfaces.
- Delivered technical workshops, product demonstrations, and customer presentations, including at Electronica 2024.
- Collaborated with applications, marketing, product, and R&D; teams to convert field requirements and issues into product feedback and improvements.

Fusionride GmbH — Senior Radar Signal Processing Engineer

Munich, Germany / India | Feb 2022 – Sept 2024

- Architected complete 1D–4D radar processing chains, including FFT, MIMO, beamforming, detection, and signal-processing modules using MATLAB/Simulink.
- Designed multi-core system architecture with focus on memory mapping, computational efficiency, runtime performance, and system integration.
- Developed antenna evaluation tools covering beam patterns, sidelobes, and virtual-array analysis, reducing analysis time from ~1 week to 1 day.
- Delivered the first 4×4 corner radar prototype with real-time detections within one year and contributed to 6×8 front-radar development.
- Led technical decisions covering algorithm architecture, system integration, validation, and performance optimization.

Continental Automotive — Technical Specialist, Radar
Bengaluru, India | Jul 2018 – Feb 2022

- Developed and validated radar signal-processing modules for automotive FMCW radar systems, achieving 99.92% functional coverage for Gen5 radar validation.
- Implemented and optimized CFAR, elevation MIMO, sidelobe suppression, and noise-estimation algorithms.
- Developed MATLAB and Embedded C modules for radar signal-processing pipelines and embedded implementation.
- Modelled and validated pulsed-Doppler radar systems and performed system-level performance analysis and debugging.

Wavelet Technologies — Project Engineer
Pune, India | Jul 2015 – Jul 2018

- Developed embedded firmware and signal-processing functionality for pulse wind-profile radar receiver systems.
- Participated in algorithm implementation, embedded development, laboratory testing, debugging, and system integration.

EDUCATION

- M.E. — Embedded Systems & VLSI | Pune University | 2016 | CGPA: 8.54/10
- B.E. — Electronics & Telecommunication | Pune University | 2013 | CGPA: 8.43/10

PUBLICATIONS & ACHIEVEMENTS

- IEEE Conference Paper: Comparative study and implementation of wind-profiler radar, 2017.
- Pioneered introduction of the first 4×4 corner radar product with real-time detections within one year at Fusionride.
- Received a monetary award for developing an engineering tool that significantly accelerated initial radar signal-processing bring-up and analysis.

LANGUAGES

English | German | Hindi | Marathi